

# Motor Datasheet: 2026-07-21

## 14-54-04 output

Operating Point: Continuous | Temperature: 120 °C

### Geometry (active elements only)

| Parameter                        | Value  | Unit |
|----------------------------------|--------|------|
| Machine Outer Diameter           | 55.0   | mm   |
| Machine Inner Diameter           | 20.0   | mm   |
| Machine Thickness / Width        | 23.7   | mm   |
| Stator Height                    | 23.7   | mm   |
| Magnet Height                    | 3.7    | mm   |
| Tooth Height                     | 13.9   | mm   |
| Air Gap                          | 0.50   | mm   |
| Fill Factor (input)              | 0.69   | -    |
| Slot Fill Factor (calculated)    | 0.56   | -    |
| Slot Width (tooth to tooth)      | 4.292  | mm   |
| Available Coil Space             | 2.146  | mm   |
| Assumed Coil Width (simulation)  | 1.946  | mm   |
| Assumed Coil Height (simulation) | 12.356 | mm   |

### Mass Properties (active elements only)

| Parameter                       | Value | Unit |
|---------------------------------|-------|------|
| Machine Mass                    | 268.3 | g    |
| Rotor Mass                      | 60.3  | g    |
| Stator Mass                     | 208.0 | g    |
| Total Copper Mass (pure copper) | 79.2  | g    |
| Total Stator Steel Mass         | 128.7 | g    |

|                                                |       |                   |
|------------------------------------------------|-------|-------------------|
| Total Magnet Mass                              | 26.2  | g                 |
| Rotor Structure Included                       | No    | -                 |
| Rotor Moment of Inertia (active elements only) | 245.8 | g·cm <sup>2</sup> |

## Electrical Properties

| Parameter                                   | Value      | Unit  |
|---------------------------------------------|------------|-------|
| Supply Current (RMS)                        | 4.908      | A     |
| Supply Current (Peak)                       | 6.941      | A     |
| Id (Peak)                                   | -0.000     | A     |
| Iq (Peak)                                   | 6.941      | A     |
| Phase Advance Angle                         | 0.0        | °elec |
| Phase Resistance at operational temperature | 101.126    | mOhm  |
| D-Axis Inductance (Ld)                      | 218.485    | μH    |
| Q-Axis Inductance (Lq)                      | 218.146    | μH    |
| Terminal Inductance (Ltt = Ld + Lq)         | 436.631    | μH    |
| PM Flux Linkage                             | 2.335      | mWb   |
| Frequency at operating point (electrical)   | 237.500    | Hz    |
| Winding Configuration                       | wye (star) | -     |
| DC Supply Voltage                           | 9.000      | V     |
| Power Factor                                | 0.880      | -     |

## Performance

| Parameter                                  | Value  | Unit      |
|--------------------------------------------|--------|-----------|
| Torque (Continuous)                        | 120.4  | mNm       |
| Speed                                      | 2850.0 | rpm       |
| Dimensionless motor constant               | 3.0    | -         |
| BackEMF Constant (Ke, phase peak)          | 1.2    | mV/rpm    |
| Motor Constant (Km, only resistive losses) | 44.5   | mNm/√W    |
| Specific Motor Constant (Km / mass)        | 165.9  | mNm/√W/kg |
| Torque Constant (Kt, peak, no-load)        | 17.5   | mNm/A     |

|               |     |   |
|---------------|-----|---|
| Torque Ripple | 0.4 | % |
| Saturation    | 1.0 | % |

## Losses

| Parameter                           | Value | Unit |
|-------------------------------------|-------|------|
| Total Losses                        | 7.8   | W    |
| Resistive Losses                    | 7.3   | W    |
| Iron Core Losses                    | 0.5   | W    |
| Magnet Eddy Current Losses          | 0.0   | W    |
| Rotor Structure Eddy Current Losses | 0.0   | W    |
| Winding Eddy Current Losses         | 0.0   | W    |

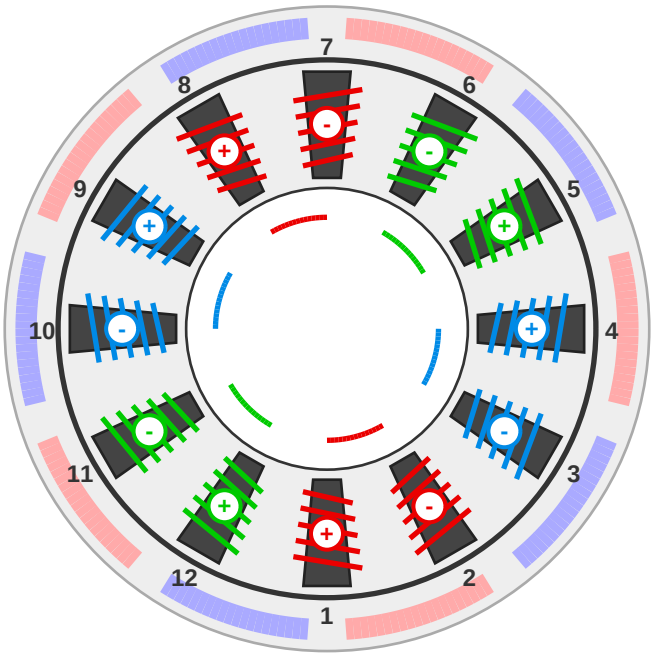
## Configuration

| Parameter                                                 | Value | Unit |
|-----------------------------------------------------------|-------|------|
| Number of Stators                                         | 1     | -    |
| Number of Rotors                                          | 1     | -    |
| Number of Slots                                           | 12    | -    |
| Pole Pairs                                                | 5     | -    |
| Number of Parallel Paths                                  | 1     | -    |
| Number of Turns per Tooth (designed incl. parallel paths) | 20    | -    |
| Number of Phase Turns (all phases in series)              | 80    | -    |

# Winding Configuration

Schema: AabBCcaABbcC

Concentrated Winding | Balanced | Cogging: 60/rev | Ripple: 60/rev



# Materials

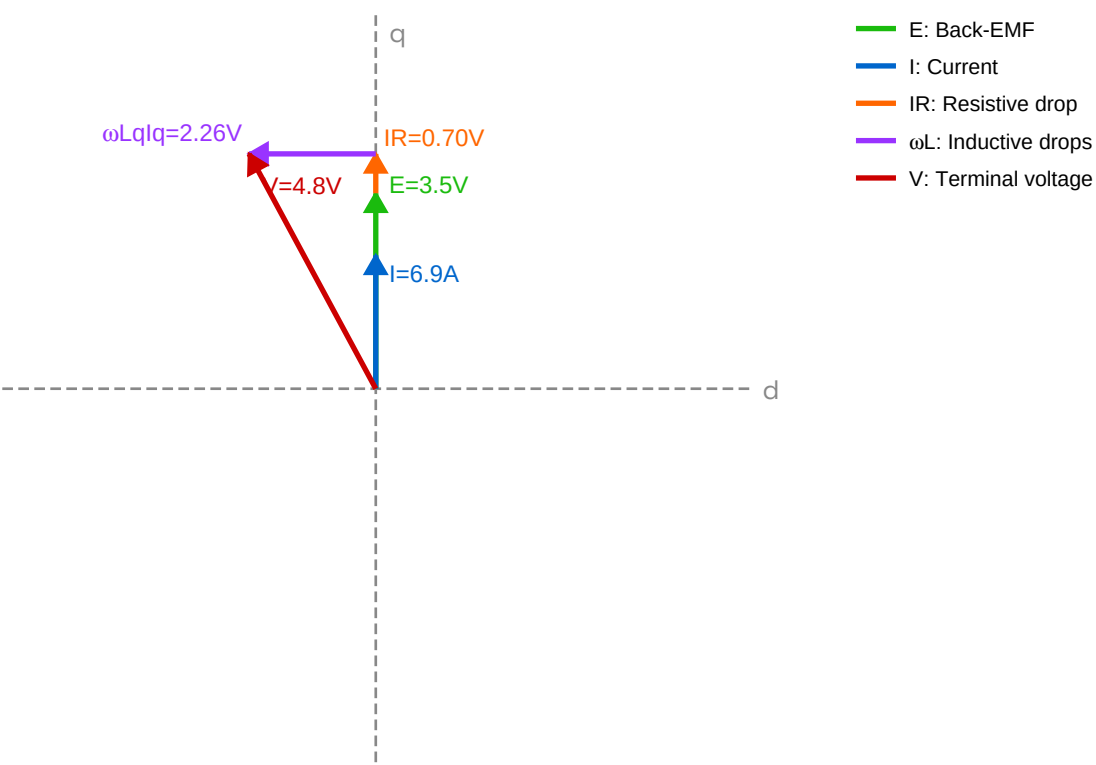
| Component      | Material                |   |
|----------------|-------------------------|---|
| Stator Core    | 700HR 5P 800MPa-Hoganas | - |
| Magnets        | BMHFa-32/27             | - |
| Rotor Backiron | 430-FR 15mm-Carpenter   | - |
| Coil           | Copper                  | - |

# Voltage Breakdown at Operating Point (with integer turns)

| Parameter                      | Value | Unit |
|--------------------------------|-------|------|
| Back-EMF Voltage (Phase, Peak) | 3.485 | V    |
| Resistive Drop (Phase, Peak)   | 0.702 | V    |
| Inductive Drop Ld (Peak)       | 0.000 | V    |
| Inductive Drop Lq (Peak)       | 2.260 | V    |

|                                       |       |   |
|---------------------------------------|-------|---|
| Inductive Drop Total (Peak)           | 2.260 | V |
| Terminal Voltage (Phase, Peak)        | 4.756 | V |
| Back-EMF Voltage (Line-to-Line, Peak) | 6.036 | V |
| Resistive Drop (Line-to-Line, Peak)   | 1.216 | V |
| Inductive Drop (Line-to-Line, Peak)   | 3.914 | V |
| Terminal Voltage (Line-to-Line, Peak) | 8.238 | V |

Phasor Diagram Phase Values NOT Line-to-Line (d-q reference frame)



## Mechanical Properties

| Parameter                           | Value  | Unit |
|-------------------------------------|--------|------|
| Airgap Axial Force (Average)        | 25.20  | N    |
| Magnet Axial Force Positive Maximum | 0.00   | N    |
| Magnet Axial Force Positive Minimum | 0.00   | N    |
| Magnet Axial Force Negative Maximum | -10.23 | N    |
| Magnet Axial Force Negative Minimum | -6.92  | N    |

## Glossary

- Peak — Refers to 0-to-peak amplitude (NOT peak-to-peak). For sinusoidal waveforms:  $\text{Peak} = \text{RMS} \times \sqrt{2}$ .
- RMS — Root Mean Square, the equivalent DC value for power calculations.
- Line-to-Line (L-L) — Voltage measured between two phases. For wye connection:  $V_{LL} = V_{\text{phase}} \times \sqrt{3}$ .